

233. Let the number be x . Then,

$$\frac{3}{4} \text{ of } \frac{4}{5} \text{ of } x - \frac{1}{6} \text{ of } \frac{2}{5} \text{ of } x = 648$$

$$\Rightarrow \frac{3x}{5} - \frac{x}{15} = 648 \Rightarrow \frac{8x}{15} = 648 \Rightarrow x = \frac{648 \times 15}{8} = 1215.$$

234. Let the required number be x .

$$\text{Then, } x + 1\frac{1}{2} = x \times 1\frac{1}{2} \Rightarrow x + \frac{3}{2} = \frac{3}{2}x \Rightarrow \frac{x}{2} = \frac{3}{2} \Rightarrow x = 3.$$

235. Let the required sum be ₹ x .

$$\text{Then, } \frac{5}{12} \text{ of } x = 3\frac{3}{4} \text{ of } 100 \Leftrightarrow \frac{5x}{12} = \left(\frac{15}{4} \times 100\right) = 375$$

$$\Leftrightarrow x = \left(\frac{375 \times 12}{5}\right) = 900.$$

236. Let x of $\frac{1}{12} = \frac{3}{8}$. Then, $\frac{x}{12} = \frac{3}{8} \Leftrightarrow x = \left(\frac{3}{8} \times 12\right) = \frac{9}{2}$.

$$\text{237. Sum of given fractions} = \frac{7}{4} + \frac{5}{2} + \frac{67}{12} + \frac{10}{3} + \frac{9}{4}$$

$$= \left(\frac{21 + 30 + 67 + 40 + 27}{12}\right) = \frac{185}{12}.$$

The whole number just less than $\frac{185}{12}$ is 15.

$$\text{Let } \frac{185}{12} - x = 15. \text{ Then, } x = \left(\frac{185}{12} - 15\right) = \frac{5}{12}.$$

$$\text{238. Sum of given fractions} = 5\frac{3}{4} + 4\frac{4}{5} + 7\frac{3}{8} = \frac{23}{4} + \frac{24}{5} + \frac{59}{8}$$

$$= \frac{230 + 192 + 295}{40} = \frac{717}{40} = 17\frac{37}{40}.$$

$$\therefore \text{ Required fraction} = \left(18 - 17\frac{37}{40}\right) = \frac{3}{40}.$$

239. Clearly, $\frac{x+1}{x}$ is the only fraction in which the numerator is greater than the denominator. So, it is the greatest fraction.

$$\text{240. } \frac{6}{7/8} - \frac{6/7}{8} = 6 \times \frac{8}{7} - \frac{6}{7} \times \frac{1}{8} = \frac{48}{7} - \frac{6}{56}$$

$$= \frac{384 - 6}{56} = \frac{378}{56} = \frac{27}{4} = 6\frac{3}{4}.$$

241. Let the value of the estate be ₹ x .

$$\text{Then, } \frac{4}{5} \text{ of } x = 16800 \Leftrightarrow x = \left(\frac{16800 \times 5}{4}\right) = 21000$$

$$\Leftrightarrow \frac{3}{7}x = \left(\frac{3}{7} \times 21000\right) = 9000.$$

242. Let the number be x . Then,

$$\frac{2}{5} \text{ of } \frac{1}{4} \text{ of } \frac{3}{7} \text{ of } x = 15 \Leftrightarrow x = \left(15 \times \frac{7}{3} \times 4 \times \frac{5}{2}\right) = 350$$

$$\Leftrightarrow \frac{1}{2}x = 175.$$

$$\text{243. Required fraction} = \frac{1 \text{ sec.}}{1 \text{ hr.}} = \frac{1 \text{ sec.}}{(1 \times 60 \times 60) \text{ sec.}} = \frac{1}{3600}.$$

$$\text{244. Height at the third bounce} = \left[32 \times \left(\frac{3}{4}\right)^3\right] \text{ m} = \left(32 \times \frac{27}{64}\right) \text{ m}$$

$$= \frac{27}{2} \text{ m} = 13\frac{1}{2} \text{ m}.$$

245. Suppose Sanket earns ₹ x in each of the other eleven months.

Then, Sanket's earning in March = ₹ $(2x)$.

Sanket's annual earning = ₹ $(11x + 2x) = ₹ (13x)$.

$$\therefore \text{ Required fraction} = \frac{2x}{13x} = \frac{2}{13}.$$

246. Let the capacity of the tank be x litres.

$$\text{Then, } \frac{1}{3}x = 80 \Leftrightarrow x = 240 \Leftrightarrow \frac{1}{2}x = 120.$$

$$\text{247. Distance travelled on foot} = \left[\frac{7}{2} - \left(\frac{5}{3} + \frac{7}{6}\right)\right] \text{ km}$$

$$= \left(\frac{7}{2} - \frac{17}{6}\right) \text{ km} = \frac{2}{3} \text{ km}.$$

$$\therefore \text{ Required fraction} = \frac{(2/3)}{(7/2)} = \left(\frac{2}{3} \times \frac{2}{7}\right) = \frac{4}{21}.$$

248. Let the required fraction be x . Then,

$$\frac{4}{7}x + \frac{4}{7} = \frac{15}{14} \Leftrightarrow \frac{4}{7}x = \left(\frac{15}{14} - \frac{4}{7}\right) = \frac{7}{14} = \frac{1}{2}$$

$$\Leftrightarrow x = \left(\frac{1}{2} \times \frac{7}{4}\right) = \frac{7}{8}.$$

$$\text{249. Required fraction} = \frac{\frac{2}{3} \text{ of } \frac{1}{4} \text{ of ₹ } 25.20}{\frac{3}{2} \text{ of ₹ } 36} = \frac{\text{₹ } 4.20}{\text{₹ } 54} = \frac{42}{540} = \frac{7}{90}.$$

250. Let the length of longer piece be x cm.

$$\text{Then, length of shorter piece} = \left(\frac{2}{5}x\right) \text{ cm}.$$

$$\therefore x + \frac{2}{5}x = 70 \Leftrightarrow \frac{7x}{5} = 70 \Leftrightarrow x = \left(\frac{70 \times 5}{7}\right) = 50.$$

$$\text{Hence, length of shorter piece} = \frac{2}{5}x = \left(\frac{2}{5} \times 50\right) \text{ cm} = 20 \text{ cm}.$$

251. Let the whole amount be ₹ x . Then,

$$A's \text{ share} = ₹ \left(\frac{3}{16}x\right); B's \text{ share} = ₹ \left(\frac{x}{4}\right);$$

$$\text{and } C's \text{ share} = ₹ \left[x - \left(\frac{3x}{16} + \frac{x}{4}\right)\right] = ₹ \left(\frac{9x}{16}\right).$$

$$\therefore \frac{9x}{16} = 81 \Leftrightarrow x = \left(\frac{81 \times 16}{9}\right) = 144.$$

$$\text{Hence, } B's \text{ share} = ₹ \left(\frac{144}{4}\right) = ₹ 36.$$

$$\text{252. Green portion} = \left[1 - \left(\frac{1}{10} + \frac{1}{20} + \frac{1}{30} + \frac{1}{40} + \frac{1}{50} + \frac{1}{60}\right)\right]$$

$$= \left[1 - \frac{1}{10} \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}\right)\right]$$